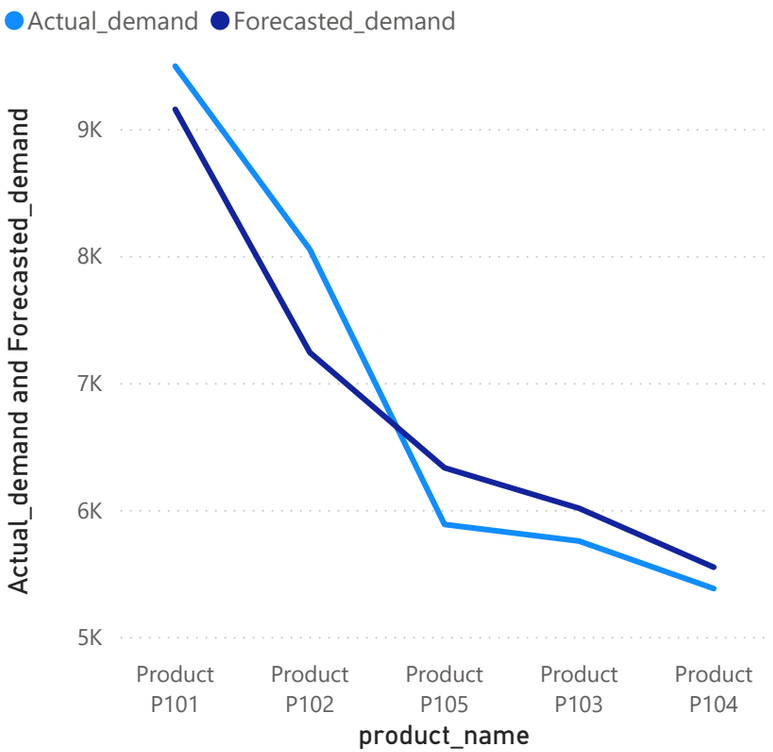
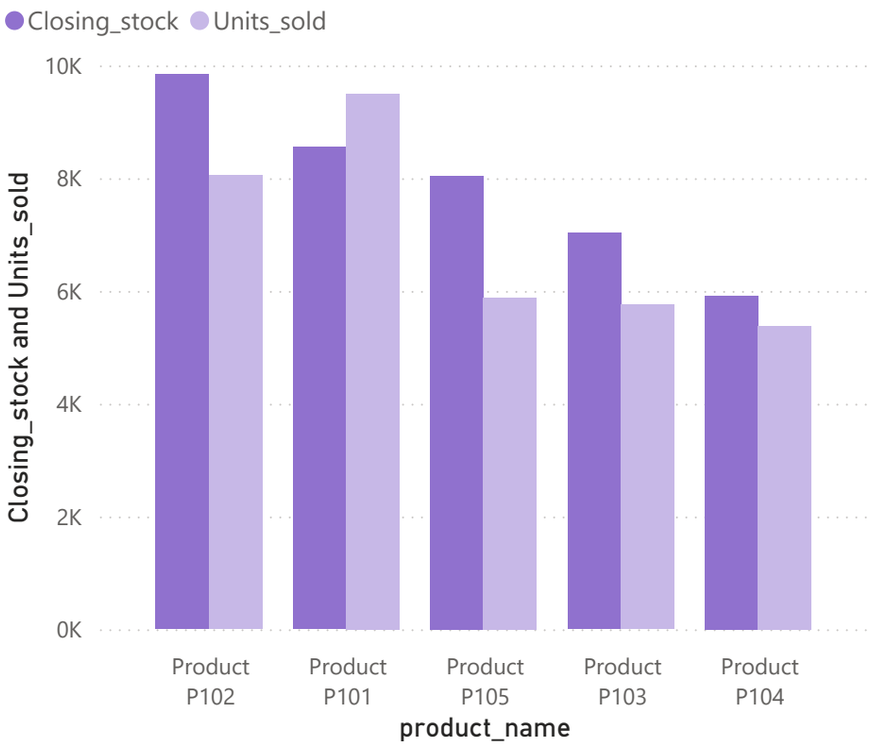
7	35K	39K	75.00%	66.99%	94.11	5.73
Stockout Count	Total Units Sold	Total Inventory	On-Time Delivery %	Forecast Accuracy %	Inventory Turnover Ratio	Average Lead Time

Forecasted vs Actual Demand



Closing_stock and Units_sold by product_name



supplier_name, product_category, product_name

- Supplier A
  - Clothing
  - Electronics
  - Home
  - Pharma
- Supplier B
- Supplier C
- Supplier D

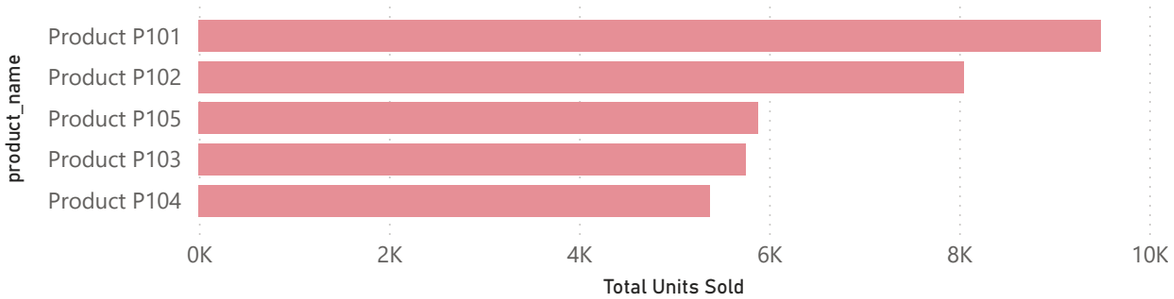
warehouse_location

- Bangalore
- Delhi
- Kolkata
- Mumbai

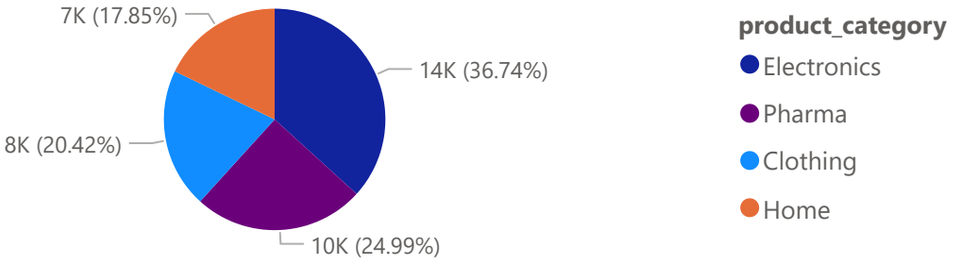
month

- 1
- 2
- 3
- 4

Total Units Sold by product_name

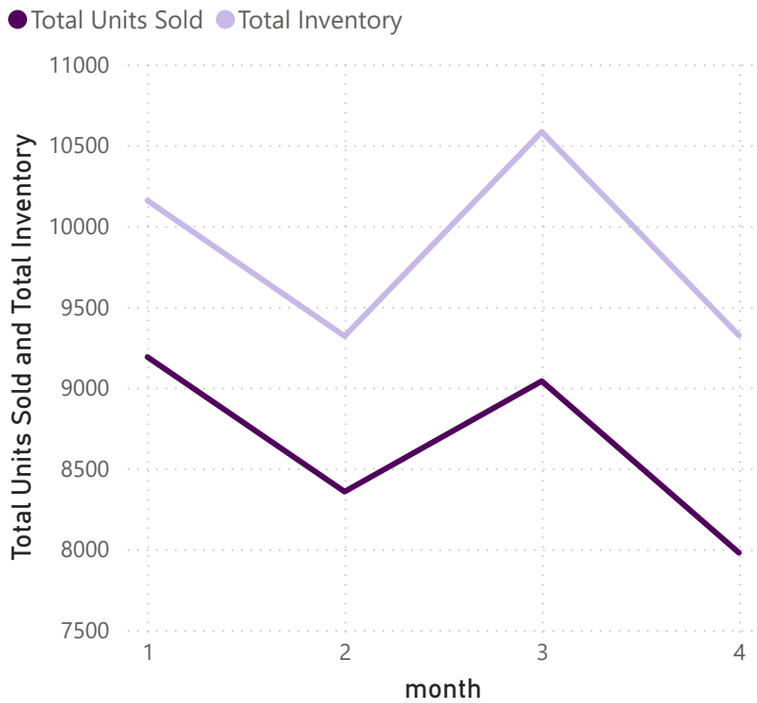


Inventory by Product Category



Total Units Sold by warehouse_location

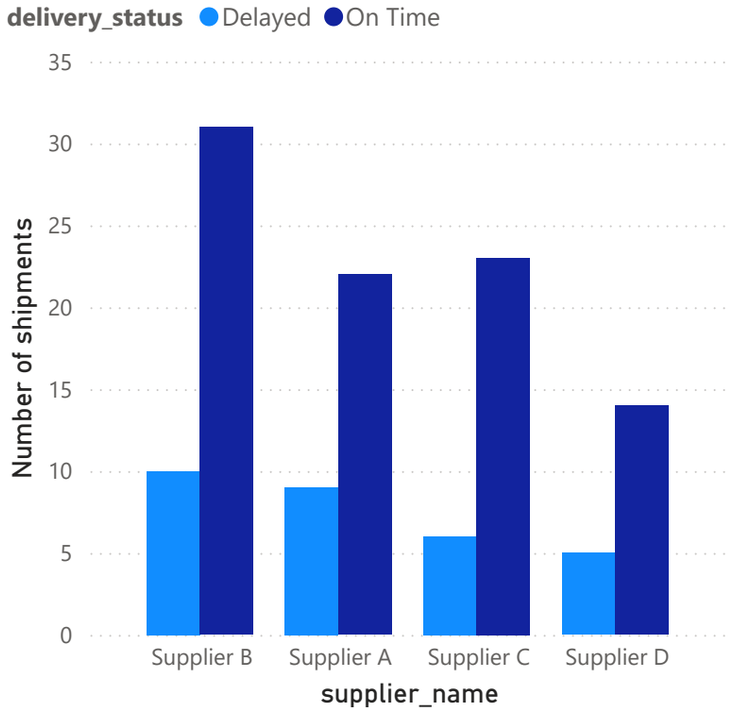
Monthly Inventory and Sales Trend



supplier_name, product_category, product_name

- Supplier A
  - Clothing
  - Electronics
  - Home
  - Pharma
- Supplier B
- Supplier C
- Supplier D

Supplier Delivery Performance



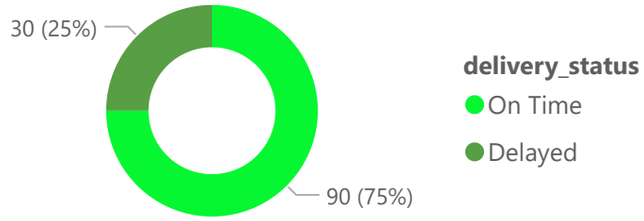
warehouse_location

- Bangalore
- Delhi
- Kolkata
- Mumbai

month

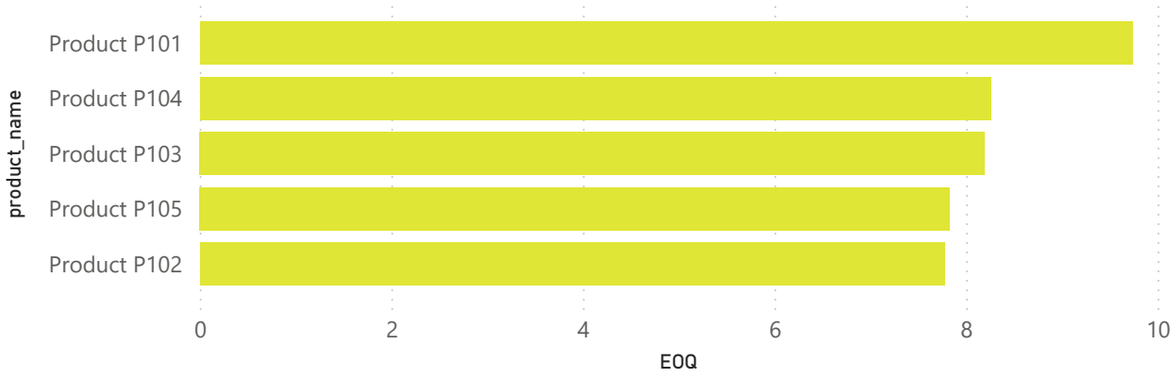
- 1
- 2
- 3
- 4

Delivery Status Distribution

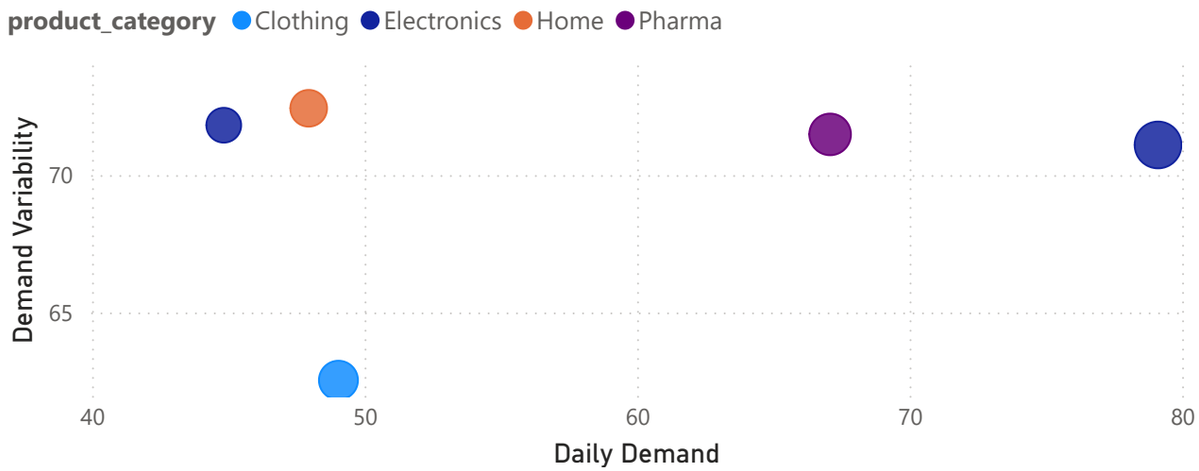


product_name	Closing_stock	Opening_stock	Reorder_level	Lead_time_days	Stockout_status	Delivery_status
Product P101	8553	11901	292	5.84	No	Delayed
Product P102	9843	11755	294	5.36	No	Delayed
Product P103	7029	8284	294	5.70	No	Delayed
Product P104	5915	7426	293	5.42	No	Delayed
Product P105	8040	9394	294	6.36	No	Delayed

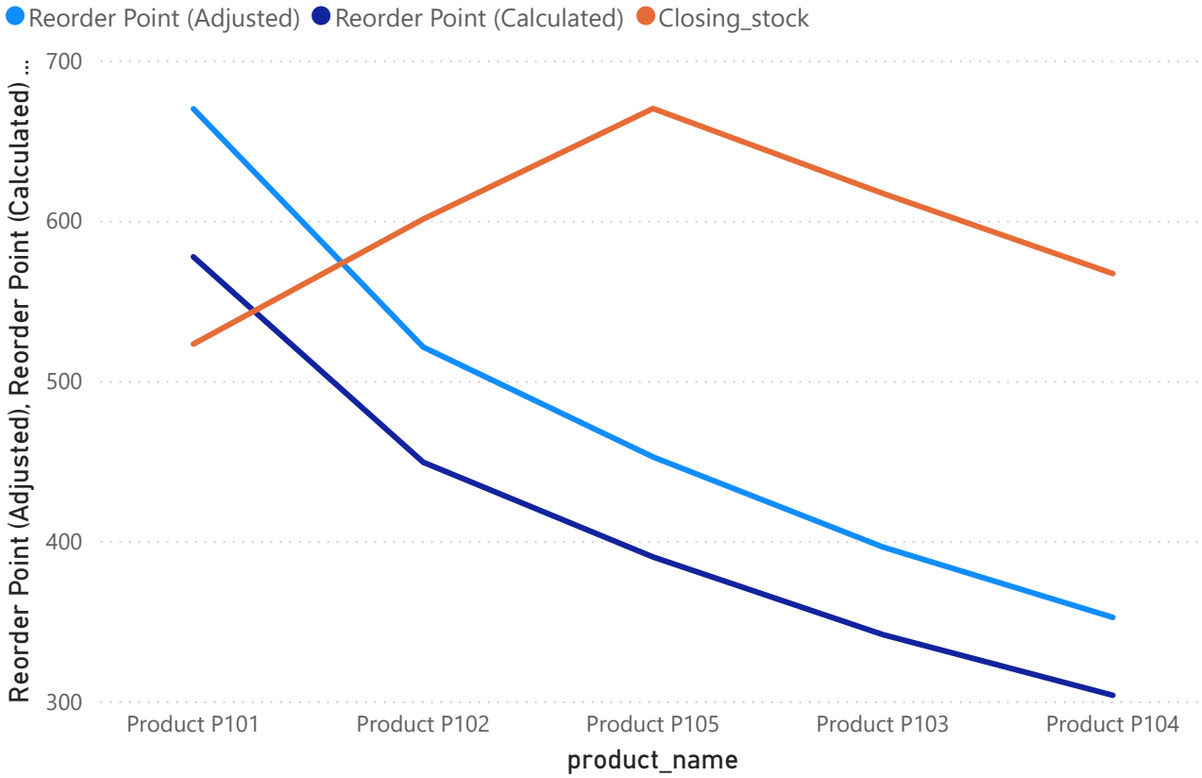
EOQ by product_name



Demand Variability vs Demand



Reorder Point (Adjusted), Reorder Point (Calculated) and Closing_stock by product_name



product_name	Safety Stock	Reorder_level (Actual)	Reorder Point (Calculated)	Stock Alert
Product P101	115.48	218.13	669.81	Reorder Needed
Product P102	89.84	222.54	521.09	Reorder Needed
Product P105	78.02	226.05	452.52	Reorder Needed
Product P103	68.33	228.20	396.31	Reorder Needed
Product P104	60.76	231.16	352.41	Reorder Needed

Lead Time Adjustment

0

Demand Growth %

0.20

Key Business Insights & Recommendations

- **Systematic Understocking Risk Identified:**
Across all products, the calculated Reorder Point (ROP) is significantly higher than the existing reorder levels. This indicates that current inventory policies underestimate demand during lead time, exposing the system to frequent stockout risks.
- **High-Risk Products Require Immediate Attention:**
Products such as P101 and P105 exhibit the highest gap between stock levels and required reorder points, making them the most vulnerable to stock shortages and service disruptions.
- **Demand Variability Drives Inventory Risk:**
The demand variability analysis shows that certain products have high fluctuations in demand. These products require higher safety stock to maintain service levels and avoid stockouts.
- **Supplier Performance Impacts Inventory Planning:**
Delays in supplier deliveries increase effective lead time, which directly raises the required reorder point and safety stock levels, impacting overall inventory efficiency.
- **Forecasting Gaps Affect Inventory Decisions:**
Mismatch between forecasted and actual demand leads to both overstocking and understocking, indicating the need for improved demand forecasting models.
- **Inventory Utilization is Uneven Across Products:**
Some products show low inventory turnover, indicating slow-moving stock and inefficient capital utilization, while others face stock shortages.
- **What-If Analysis Reveals Sensitivity to Demand & Lead Time:**
Simulation using demand growth and lead time adjustments shows that even small increases in demand or delays in lead time significantly increase the reorder point, highlighting the sensitivity of inventory decisions to operational uncertainties.
- **Reorder Policies Need Dynamic Adjustment:**
Static reorder levels are insufficient in a dynamic environment. The analysis demonstrates the need for data-driven, flexible inventory policies that adapt to demand changes and supply variability.
- **Opportunity for Supply Chain Optimization:**
By aligning reorder levels with calculated ROP, improving supplier reliability, and enhancing forecast accuracy, organizations can reduce stockouts, optimize inventory holding costs, and improve service levels.

Strategic Recommendation

Adopting a data-driven inventory management approach with scenario-based planning (what-if analysis) enables proactive decision-making and strengthens overall supply chain resilience